

CSE 717 — Information Security

Topic 13: DES / Feistel Structure / S-box / Avalanche Effect / 3-DES

Complete Past Paper Solutions: 2020–2024

Source: general security knowledge (Stallings DES) — NOT in any 2026 Lc material,
but CONFIRMED EVERY YEAR

Every Topic-13 question 2020–2024 included. Zero omissions.

Contents

1	2024 — Q6(d): Avalanche Effect + DES Attack Families (2 marks)	5
2	2023 — Q3(a,b): 3-DES + DES Equation, Avalanche (5 marks); Section B Q5(a): F-function (2.5 marks)	6
2.1	Part (a): 3-DES Block Diagram + DES Encryption Equation	6
2.2	Part (b): Avalanche Effect + Attack Families	6
2.3	Part Section B Q5(a): The F-function	6
3	2022 — Q4: S-box + 3-DES + F-function (8 marks)	7
3.1	Part (a): Why S-box + S1-Box(37)	7
3.2	Part (b): 3-DES Schematic + Encryption Equation	7
3.3	Part (c): F-function Components	7
4	2021 — A4: IP^{-1} Equation, 3-DES, Avalanche (9.5 marks)	8
4.1	Part (i): IP^{-1} Equation	8
4.2	Part (ii): 3-DES Block Diagram + Equation	8
4.3	Part (iii): Avalanche Effect + Attack Families	8
5	2020 — Q4(a,b): IP/IP^{-1}, S1-Box(37), F-function, Finite Fields (9.5 marks); Section B Q5(a,b): 3-DES + Avalanche + Feistel (8.25 marks)	9
5.1	Part Q4(a): IP/IP^{-1} Purpose + S1-Box(37)	9
5.2	Part Q4(b): IP^{-1} Decryption Equation + Finite Fields	9
5.3	Part Section B Q5(a): 3-DES Block Diagram + Equation	10
5.4	Part Section B Q5(b): Avalanche + Attack Families + Classical Feistel Structure	10

Quick Reference — The 6 Recurring Sub-Patterns (CHEAT SHEET)

Not in any 2026 Lc material — **CONFIRMED GUARANTEED** every year (5/5, 5–10)

None of Lc#1–11 cover DES/Feistel/3-DES content. Everything below is standard textbook (Stallings) knowledge memorized as a cheat sheet — 45min investment for 6–10 guaranteed marks.

1. Classical Feistel Cipher Structure

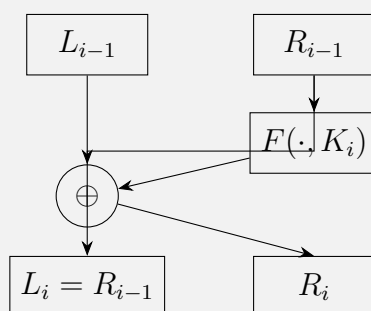
1. Split the plaintext block into two equal halves: L_0 (left) and R_0 (right).
2. For each round $i = 1, \dots, n$ (DES uses $n = 16$):

$$L_i = R_{i-1}, \quad R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

where F is the round (mangler) function and K_i is the i -th round subkey.

3. After the final round, the two halves are **combined** (and typically swapped back) to form the ciphertext block.

One-round diagram:



Key property: the SAME algorithm performs both encryption and decryption — decryption simply applies the subkeys $K_n, \dots, K_1$ in **reverse order**. This symmetry is the entire point of a Feistel network.

2. DES Overview + IP / IP^{-1} Permutations

- **DES parameters:** 64-bit plaintext block, 56-bit effective key (64 bits incl. parity), **16 Feistel rounds**.
- **IP (Initial Permutation):** reorders the 64 input bits *before* round 1. **Purpose:** purely a bit-level rearrangement for convenient loading into 1970s hardware registers — has **no cryptographic significance** on its own.
- **IP^{-1} (Final/Inverse Permutation):** applied *after* round 16 to undo IP's bit-ordering and produce the 64-bit ciphertext:

$$\text{Ciphertext} = IP^{-1}(R_{16} \parallel L_{16})$$

(note the swap of the two halves before the final permutation — this swap is what makes the SAME structure work for decryption).

3. F-function (Round/Mangler Function) Components

Input: R_{i-1} (32 bits) and round subkey K_i (48 bits). Steps:

1. **Expansion permutation E :** expands R_{i-1} from 32 $\rightarrow$ 48 bits (some bits duplicated).
2. **XOR with round key:** $E(R_{i-1}) \oplus K_i$ (48 bits).
3. **S-box substitution:** the 48 bits are split into eight 6-bit groups, each passed through one of **8 S-boxes** (S_1 – S_8), each producing 4 bits $\rightarrow$ 32 bits total. **This is the only non-linear step** — the core source of DES's security (confusion).
4. **Permutation P :** a 32 $\rightarrow$ 32-bit straight permutation that shuffles the S-box outputs (diffusion).

$$F(R_{i-1}, K_i) = P\left(S(E(R_{i-1}) \oplus K_i)\right)$$

Diagram (chain form):

$$R_{i-1} \xrightarrow{E(32 \rightarrow 48)} \boxed{\oplus K_i} \rightarrow \boxed{S_1 \dots S_8 (48 \rightarrow 32)} \xrightarrow{P(32 \rightarrow 32)} F(R_{i-1}, K_i)$$

4. S-box / S1-Box Lookup for Input 37 (recurs 2020 & 2022)

Why S-boxes exist: they are the **only non-linear component** of DES — without them, DES would be a linear (and trivially breakable) cipher. S-boxes provide **confusion**.

Lookup procedure (any 6-bit input $b_1b_2b_3b_4b_5b_6$):

- **Row** = b_1b_6 (outer bits, as a 2-bit binary number, 0–3).
- **Column** = $b_2b_3b_4b_5$ (middle 4 bits, as a 4-bit binary number, 0–15).
- Look up $S_1[\text{row}][\text{column}]$ in the standard S_1 table.

Worked example — input 37:

- $37 = 100101_2$ (6 bits).
- Row = $b_1b_6 = 11_2 = 3$. Column = $b_2b_3b_4b_5 = 0010_2 = 2$.
- S_1 table, row 3: 15 12 8 2 4 9 1 7 5 11 3 14 10 0 6 13 (columns 0–15).
- Column 2 $\rightarrow$ value = 8 (binary 1000).

$$\boxed{S_1(37) = 8}$$

5. Avalanche Effect + Two Families of DES Attacks (recurs EVERY year)

- **Avalanche effect:** a small change in the input (even 1 bit of plaintext or key) should produce a **large, unpredictable change** in the output ciphertext (ideally $\sim$ half the output bits flip). A strong avalanche effect makes statistical attacks hard.
- **Two families of attacks on DES:**
 1. **Differential cryptanalysis** (Biham & Shamir) — studies how differences in plaintext pairs propagate to differences in ciphertext pairs to

recover key bits.

2. **Linear cryptanalysis** (Matsui) — finds linear approximations relating plaintext, ciphertext, and key bits that hold with probability $\neq 0.5$, exploited statistically over many samples.

6. 3-DES (Triple DES) Block Diagram + Encryption Equation

Why 3-DES: single DES's 56-bit key is too short (brute-forceable). 3-DES applies DES three times with (typically 2 or 3) different keys to extend effective key length, while staying **backward-compatible** with single DES.

Encryption (EDE = Encrypt-Decrypt-Encrypt):

$$C = E_{K_3}(D_{K_2}(E_{K_1}(P)))$$

Decryption:

$$P = D_{K_1}(E_{K_2}(D_{K_3}(C)))$$

Block diagram (chain form):

$$P \xrightarrow{E_{K_1}} \boxed{\text{stage 1}} \xrightarrow{D_{K_2}} \boxed{\text{stage 2}} \xrightarrow{E_{K_3}} \boxed{\text{stage 3}} \rightarrow C$$

Why EDE (not EEE), and why a middle *decrypt*: if $K_1 = K_2 = K_3$, then $D_{K_2}(E_{K_1}(P)) = P$, so $C = E_{K_3}(P)$ — i.e. 3-DES with all-equal keys collapses to **plain single DES**, giving **backward compatibility** with legacy single-DES systems. With 2 distinct keys ($K_1 = K_3 \neq K_2$), effective key length ≈ 112 bits; with 3 distinct keys, ≈ 168 bits.

Bonus: Finite Fields in Cryptography (one-liner, 2020 Q4(b))

A **finite field** $GF(2^n)$ is the set of all n -bit values with addition (bitwise XOR) and multiplication defined modulo an **irreducible polynomial** of degree n . This gives every non-zero element a multiplicative inverse — the algebraic structure needed for **invertible** (hence decryptable) substitution operations. Used directly in AES ($GF(2^8)$, S-box & MixColumns).

2024 — Q6(d): Avalanche Effect + DES Attack Families (2 marks)

Question — 2024 Q6(d)

Explain the avalanche effect in the context of DES, and name the two families of attacks against DES. [2]

Answer — 2024 Q6(d)

- **Avalanche effect:** a 1-bit change in the plaintext or key causes a large, unpredictable change in the ciphertext — roughly half the output bits flip. This prevents an attacker from inferring small input changes from small output changes.
- **Two attack families:** **Differential cryptanalysis** (analyzes how input differences propagate to output differences) and **Linear cryptanalysis** (exploits linear approximations of the cipher that hold with non-trivial probability).

See Cheat Sheet #5 for full definitions.

2023 — Q3(a,b): 3-DES + DES Equation, Avalanche (5 marks); Section B Q5(a): F-function (2.5 marks)

Question — 2023 Q3

- (a) Draw the 3-DES block diagram and write the DES encryption equation. [3]
- (b) Explain the avalanche effect and name the two families of attacks against DES. [2]

Section B Q5(a): Describe the F-function (round function) used in DES. [2.5]

Part (a): 3-DES Block Diagram + DES Encryption Equation

Answer — 2023 Q3(a)

3-DES block diagram: see Cheat Sheet #6 — $P \xrightarrow{E_{K_1}} \xrightarrow{D_{K_2}} \xrightarrow{E_{K_3}} C$.
Single-DES encryption equation (per round):

$$L_i = R_{i-1}, \quad R_i = L_{i-1} \oplus F(R_{i-1}, K_i), \quad i = 1, \dots, 16$$

$$\text{Ciphertext} = IP^{-1}(R_{16} \parallel L_{16})$$

Part (b): Avalanche Effect + Attack Families

Answer — 2023 Q3(b)

Same as 2024 Q6(d) above — see Cheat Sheet #5: **avalanche effect** (1-bit change → large unpredictable ciphertext change) + **differential** and **linear cryptanalysis**.

Part Section B Q5(a): The F-function

Answer — 2023 Section B Q5(a)

The **F-function** (round/mangler function) takes R_{i-1} (32 bits) and subkey K_i (48 bits), and produces a 32-bit output via 4 stages:

1. **Expansion (E):** $32 \rightarrow 48$ bits (some bits duplicated).
2. **XOR with K_i :** $E(R_{i-1}) \oplus K_i$.
3. **S-box substitution:** eight 6-bit groups → eight S-boxes (S_1 – S_8) → 32 bits (the only non-linear step — confusion).
4. **Permutation (P):** $32 \rightarrow 32$ -bit shuffle (diffusion).

$$F(R_{i-1}, K_i) = P(S(E(R_{i-1}) \oplus K_i))$$

See Cheat Sheet #3 for the diagram.

2022 — Q4: S-box + 3-DES + F-function (8 marks)

Question — 2022 Q4

- (a) Why is an S-box needed in DES? Given input value 37, find the corresponding S_1 -box output. [2]
- (b) Draw the 3-DES schematic and write the encryption equation. [3]
- (c) Describe the components of the f-function (F-function) in DES. [3]

Part (a): Why S-box + S1-Box(37)

Answer — 2022 Q4(a)

Why S-box: the S-box is the **only non-linear element** in DES's round function — it provides **confusion** (obscures the relationship between key and ciphertext). Without it, DES would be a purely linear transformation and trivially breakable by solving linear equations.

S_1 -box lookup for input 37:

- $37 = 100101_2$. Row = $b_1b_6 = 11_2 = 3$, Column = $b_2b_3b_4b_5 = 0010_2 = 2$.
- S_1 row 3: 15 12 8 2 4 9 1 7 5 11 3 14 10 0 6 13 $\rightarrow$ column 2 = 8.

$$S_1(37) = 8$$

Part (b): 3-DES Schematic + Encryption Equation

Answer — 2022 Q4(b)

See Cheat Sheet #6:

$$P \xrightarrow{E_{K_1}} \boxed{\text{stage 1}} \xrightarrow{D_{K_2}} \boxed{\text{stage 2}} \xrightarrow{E_{K_3}} \boxed{\text{stage 3}} \rightarrow C \quad C = E_{K_3}(D_{K_2}(E_{K_1}(P)))$$

With $K_1 = K_2 = K_3$, this collapses to single DES (backward compatibility). 2-key 3-DES ($K_1 = K_3$) gives ≈ 112 -bit effective security; 3-key gives ≈ 168 -bit.

Part (c): F-function Components

Answer — 2022 Q4(c)

Same 4-stage description as 2023 Section B Q5(a) above: **Expansion (32 $\rightarrow$ 48) $\rightarrow$ XOR with $K_i \rightarrow$ S-box substitution (48 $\rightarrow$ 32, 8 S-boxes) $\rightarrow$ Permutation P (32 $\rightarrow$ 32). See Cheat Sheet #3 for the diagram.**

2021 — A4: IP^{-1} Equation, 3-DES, Avalanche (9.5 marks)

Question — 2021 A4

- (i) Explain the IP^{-1} (inverse initial permutation) equation in DES. [part of 4.75]
- (ii) Draw the 3-DES block diagram and write its encryption equation. [2.75]
- (iii) Explain the avalanche effect and name the two families of attacks against DES. [2]

Part (i): IP^{-1} Equation

Answer — 2021 A4(i)

After 16 Feistel rounds, the two 32-bit halves are **swapped** and the inverse initial permutation IP^{-1} is applied to produce the final 64-bit ciphertext:

$$\text{Ciphertext} = IP^{-1}(R_{16} \parallel L_{16})$$

IP^{-1} exactly **undoes** the bit-reordering performed by IP at the start — so $IP^{-1}(IP(X)) = X$. Neither IP nor IP^{-1} adds cryptographic strength; they exist only for hardware bit-ordering convenience (see Cheat Sheet #2).

Part (ii): 3-DES Block Diagram + Equation

Answer — 2021 A4(ii)

Identical to Cheat Sheet #6: $P \xrightarrow{E_{K_1}} \xrightarrow{D_{K_2}} \xrightarrow{E_{K_3}} C$, with $C = E_{K_3}(D_{K_2}(E_{K_1}(P)))$.

Part (iii): Avalanche Effect + Attack Families

Answer — 2021 A4(iii)

Same as Cheat Sheet #5: **avalanche effect** + **differential cryptanalysis** & **linear cryptanalysis**.

2020 — Q4(a,b): IP/IP⁻¹, S1-Box(37), F-function, Finite Fields (9.5 marks); Section B Q5(a,b): 3-DES + Avalanche + Feistel (8.25 marks)

Question — 2020 Q4 and Section B Q5

Q4.

- (a) Explain the purpose of IP and IP^{-1} in DES. Given input 37, find the S_1 -box output. [2 + 2.75]
- (b) Write the IP^{-1} decryption equation. Briefly explain finite fields in cryptography. [part of total]

Section B Q5.

- (a) Draw the 3-DES block diagram and write its encryption equation (twice, 2.75×2). [5.5]
- (b) Explain the avalanche effect, the two families of attacks against DES, and the classical Feistel cipher structure. [2.75]

Part Q4(a): IP/IP⁻¹ Purpose + S1-Box(37)

Answer — 2020 Q4(a)

Purpose of IP / IP^{-1} : IP reorders the 64 input bits before round 1; IP^{-1} undoes this reordering after round 16 (applied to $R_{16}||L_{16}$) to produce the ciphertext. **Neither adds cryptographic strength** — purely a hardware bit-loading convenience (see Cheat Sheet #2).

S_1 -box(37): same computation as Cheat Sheet #4 → $S_1(37) = 8$.

Part Q4(b): IP⁻¹ Decryption Equation + Finite Fields

Answer — 2020 Q4(b)

IP^{-1} decryption equation: same formula used in both directions — Output = $IP^{-1}(R_{16}||L_{16})$. In decryption, the ciphertext is fed back through the same 16-round structure with subkeys in reverse order ($K_{16}, \dots, K_1$); IP and IP^{-1} are applied identically since $IP^{-1} = IP^{-1}$ regardless of direction.

Finite fields in cryptography: see Cheat Sheet (Bonus) — $GF(2^n)$, XOR addition, multiplication mod an irreducible polynomial, gives every element a multiplicative inverse (needed for invertible operations). Used in AES's $GF(2^8)$.

Part Section B Q5(a): 3-DES Block Diagram + Equation

Answer — 2020 Section B Q5(a)

Same as Cheat Sheet #6: $P \xrightarrow{E_{K_1}} \xrightarrow{D_{K_2}} \xrightarrow{E_{K_3}} C$, $C = E_{K_3}(D_{K_2}(E_{K_1}(P)))$, decryption $P = D_{K_1}(E_{K_2}(D_{K_3}(C)))$.

Part Section B Q5(b): Avalanche + Attack Families + Classical Feistel Structure

Answer — 2020 Section B Q5(b)

- **Avalanche effect + 2 attack families:** see Cheat Sheet #5 (differential & linear cryptanalysis).
- **Classical Feistel structure:** see Cheat Sheet #1 — split block into L_0, R_0 ; each round $L_i = R_{i-1}$, $R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$; decryption uses the SAME structure with subkeys reversed.

Exam Strategy Summary

High-Yield Checklist for DES/Feistel/3-DES (5/5 years, 5–10 marks — GUARANTEED)

1. **3-DES block diagram + encryption equation** $C = E_{K_3}(D_{K_2}(E_{K_1}(P)))$ — asked EVERY year (2020–2024). Memorize the EDE chain and WHY (backward compatibility with single DES when $K_1 = K_2 = K_3$).
2. **Avalanche effect + 2 attack families** (differential & linear cryptanalysis) — asked EVERY year. One-liner each.
3. **F-function 4 stages:** Expansion (32→48) → XOR with K_i → S-box (48→32, only non-linear step) → Permutation P (32→32). Asked 2022, 2023.
4. **S1-box lookup for input 37:** row= b_1b_6 , col= $b_2b_3b_4b_5$ → row 3, col 2 → 8. IDENTICAL in 2020 & 2022 — memorize this one worked example.
5. **IP/IP⁻¹ purpose:** bit-reordering only, no cryptographic strength; IP^{-1} applied to $R_{16}||L_{16}$ (note the swap). Asked 2020, 2021.
6. **Classical Feistel structure:** $L_i = R_{i-1}$, $R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$; same algorithm for encrypt/decrypt with reversed subkeys. Asked 2020.
7. **Bonus — finite fields ($GF(2^n)$):** one-liner, asked once (2020), low effort.

Total: 45 minutes of focused memorization for 6–10 GUARANTEED marks every single year.